

Двойной стек IPv4 и IPv6 в GNS3

Лабораторная работа №6

Суннатилло Махмудов

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Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Изучение принципов настройки адресного пространства IPv4 и IPv6, построение сети с двойным стеком и проверка взаимодействия устройств в среде **GNS3**.

1. Выполнить разбиение IPv4 и IPv6 сетей на подсети.
2. Настроить адресацию на маршрутизаторах **FRR** и **VyOS**.
3. Проверить связность между узлами по протоколам **ICMP** и **ICMPv6**.
4. Проанализировать трафик с помощью **Wireshark**.
5. Подтвердить корректность работы протоколов **ARP**, **ICMP** и **ICMPv6**.

Теоретическая часть

- **GNS3** — эмулятор сетевой инфраструктуры.
- **FRRouting (FRR)** — платформа маршрутизации уровня L3.
- **VyOS** — дистрибутив Linux с функциями маршрутизатора.
- **VPCS** — эмулятор конечных узлов.
- **Wireshark** — анализатор сетевого трафика.

Разбиение сетей

- Разбита на три подсети:
 - /25 — 126 узлов
 - /26 — 62 узла
 - /26 — 62 узла
- Broadcast-адреса: 172.16.20.127, 172.16.20.191, 172.16.20.255.

- Разделена на две подсети /49 по идентификатору подсети.
- Вариант деления по Interface ID — два блока /65.

Моделирование сети

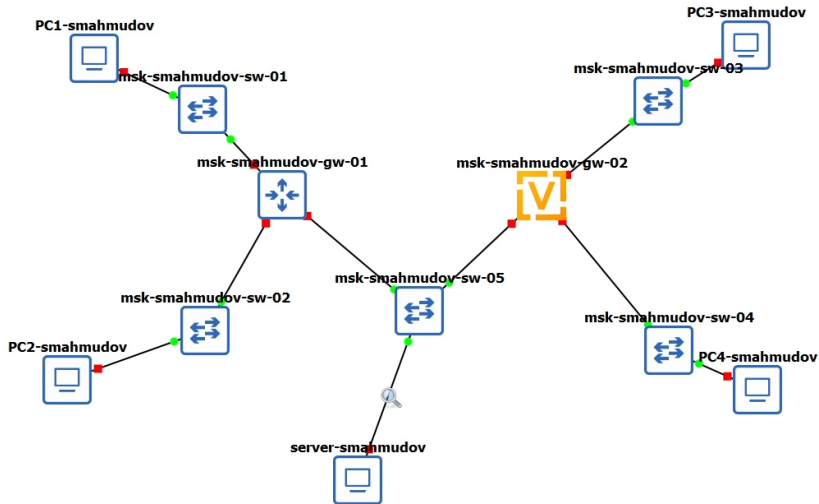


Рис. 1: Топология сети

```
PC1-smahmudov - PuTTY
VPCS> show ip
NAME           : VPCS[1]
IP/MASK        : 172.16.20.10/25
GATEWAY        : 172.16.20.1
DNS            :
MAC            : 00:50:79:66:68:00
LPORT          : 10022
RHOST:PORT     : 127.0.0.1:10023
MTU            : 1500

VPCS> show ipv6
NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    :
DNS            :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:00
LPORT          : 10022
RHOST:PORT     : 127.0.0.1:10023
MTU            : 1500

VPCS> █
```

```
msk-smahmudov-gw-01 - PuTTY

Hello, this is FRRouting (version 8.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

frr# configure terminal
frr(config)# hostname msk-smahmudov-gw-01
msk-smahmudov-gw-01(config)# exit
msk-smahmudov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-smahmudov-gw-01# configure terminal
msk-smahmudov-gw-01(config)# interface eth0
msk-smahmudov-gw-01(config-if)# ip address 172.16.20.1/25
msk-smahmudov-gw-01(config-if)# no shutdown
msk-smahmudov-gw-01(config-if)# exit
msk-smahmudov-gw-01(config)# interface eth1
msk-smahmudov-gw-01(config-if)# ip address 172.16.20.129/25
msk-smahmudov-gw-01(config-if)# no shutdown
msk-smahmudov-gw-01(config-if)# exit
msk-smahmudov-gw-01(config)# interface eth2
msk-smahmudov-gw-01(config-if)# ip address 64.100.1.1/24
msk-smahmudov-gw-01(config-if)# no shutdown
msk-smahmudov-gw-01(config-if)# exit
msk-smahmudov-gw-01(config)# exit
msk-smahmudov-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-smahmudov-gw-01#
```

The image shows two overlapping PuTTY terminal windows. The left window, titled 'PC1-smahmudov - PuTTY', shows a trace to 172.16.20.10 and a ping to 64.100.1.10. The right window, titled 'PC2-smahmudov - PuTTY', shows a ping to 172.16.20.10 and a trace to 64.100.1.10. Both windows show successful ping results with 84 bytes, TTL=63, and various response times.

```
PC1-smahmudov - PuTTY
84 bytes from 172.16.20.138 icmp_seq=3 ttl=63 time=1.659 ms
84 bytes from 172.16.20.138 icmp_seq=4 ttl=63 time=3.381 ms
84 bytes from 172.16.20.138 icmp_seq=5 ttl=63 time=2.510 ms

VPCS> trace 172.16.20.10
trace to 172.16.20.10, 8 hops
 1 172.16.20.1 3.68
 2 *172.16.20.138 2
e)
VPCS> trace 64.100.1.10
trace to 64.100.1.10, 8 hops
 1 172.16.20.1 1.46
 2 *64.100.1.10 1.3
VPCS> ping 64.100.1.10
84 bytes from 64.100.1.10 icmp_seq=1 ttl=63 time=2.850 ms
84 bytes from 64.100.1.10 icmp_seq=2 ttl=63 time=2.086 ms
84 bytes from 64.100.1.10 icmp_seq=3 ttl=63 time=1.772 ms
84 bytes from 64.100.1.10 icmp_seq=4 ttl=63 time=3.808 ms
84 bytes from 64.100.1.10 icmp_seq=5 ttl=63 time=2.748 ms
VPCS>

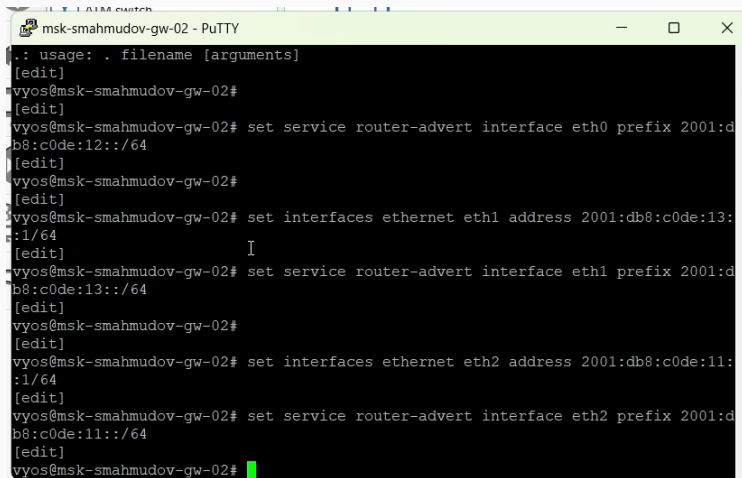
PC2-smahmudov - PuTTY
VPCS> ping 172.16.20.10
84 bytes from 172.16.20.10 icmp_seq=1 ttl=63 time=7.232 ms
84 bytes from 172.16.20.10 icmp_seq=2 ttl=63 time=3.333 ms
84 bytes from 172.16.20.10 icmp_seq=3 ttl=63 time=3.484 ms
84 bytes from 172.16.20.10 icmp_seq=4 ttl=63 time=2.025 ms
84 bytes from 172.16.20.10 icmp_seq=5 ttl=63 time=1.531 ms
VPCS> ping 64.100.1.10
84 bytes from 64.100.1.10 icmp_seq=1 ttl=63 time=2.850 ms
84 bytes from 64.100.1.10 icmp_seq=2 ttl=63 time=2.086 ms
84 bytes from 64.100.1.10 icmp_seq=3 ttl=63 time=1.772 ms
84 bytes from 64.100.1.10 icmp_seq=4 ttl=63 time=3.808 ms
84 bytes from 64.100.1.10 icmp_seq=5 ttl=63 time=2.748 ms
VPCS>
```

Рис. 4: Ping IPv4

```
server-smahmudov - PuTTY
VPCS> show ip
NAME           : VPCS[1]
IP/MASK        : 64.100.1.10/24
GATEWAY        : 64.100.1.1
DNS            :
MAC            : 00:50:79:66:68:04
LPORT         : 10026
RHOST:PORT     : 127.0.0.1:10027
MTU            : 1500

VPCS> show ipv6
NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6804/64
GLOBAL SCOPE    : 2001:db8:c0de:11::a/64
DNS            :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:04
LPORT         : 10026
RHOST:PORT     : 127.0.0.1:10027
MTU            : 1500

VPCS>
```



```
msk-smahmudov-gw-02 - PuTTY
.: usage: . filename [arguments]
[edit]
vyos@msk-smahmudov-gw-02#
[edit]
vyos@msk-smahmudov-gw-02# set service router-advert interface eth0 prefix 2001:d
b8:c0de:12::/64
[edit]
vyos@msk-smahmudov-gw-02#
[edit]
vyos@msk-smahmudov-gw-02# set interfaces ethernet eth1 address 2001:db8:c0de:13:
:1/64
[edit]
vyos@msk-smahmudov-gw-02# set service router-advert interface eth1 prefix 2001:d
b8:c0de:13::/64
[edit]
vyos@msk-smahmudov-gw-02#
[edit]
vyos@msk-smahmudov-gw-02# set interfaces ethernet eth2 address 2001:db8:c0de:11:
:1/64
[edit]
vyos@msk-smahmudov-gw-02# set service router-advert interface eth2 prefix 2001:d
b8:c0de:11::/64
[edit]
vyos@msk-smahmudov-gw-02#
```

Рис. 6: VyOS конфигурация


```
PC3-smahmudov - PuTTY
2001:db8:c0de:13::a icmp6_seq=3 ttl=62 time=1.998 ms
2001:db8:c0de:13::a icmp6_seq=4 ttl=62 time=3.912 ms
2001:db8:c0de:13::a icmp6_seq=5 ttl=62 time=2.161 ms

VPCS> ping 2001:db8:c0de:11::a

2001:db8:c0de:11::a icmp6_seq=1 ttl=62 time=4.322 ms
2001:db8:c0de:11::a icmp6_seq=2 ttl=62 time=3.783 ms
2001:db8:c0de:11::a icmp6_seq=3 ttl=62 time=2.252 ms
2001:db8:c0de:11::a icmp6_seq=4 ttl=62 time=2.666 ms
2001:db8:c0de:11::a icmp6_seq=5 ttl=62 time=2.938 ms

VPCS> trace 2001:db8:c0de:11::a

trace to 2001:db8:c0de:11::a, 64 hops max
 1 2001:db8:c0de:12::1    2.112 ms  1.263 ms  1.549 ms
 2 2001:db8:c0de:11::a    2.117 ms  1.640 ms  1.343 ms

VPCS>
VPCS> ping 172.16.20.138

host (172.16.20.138) not reachable

VPCS>
```

The image shows the Wireshark network traffic analysis interface. The top pane displays a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The bottom pane shows the detailed view of the selected packet (No. 33), which is an ARP request.

No.	Time	Source	Destination	Protocol	Length	Info
30	96.349948	64.100.1.10	172.16.20.138	ICMP	98	Echo (ping) request id=0xf780, seq=5/1280, ttl=6
31	96.352160	172.16.20.138	64.100.1.10	ICMP	98	Echo (ping) reply id=0xf780, seq=5/1280, ttl=6
32	97.383319	0c:dc:1d:6f:00:02	Private_66:68:04	ARP	60	Who has 64.100.1.10? Tell 64.100.1.1
33	97.383621	Private_66:68:04	0c:dc:1d:6f:00:02	ARP	60	64.100.1.10 is at 00:50:79:66:68:04
34	116.636614	2001:db8:c0de:11::a	2001:db8:c0de:13::a	ICMPv6	118	Echo (ping) request id=0x0c81, seq=1, hop limit=6
35	116.640169	2001:db8:c0de:13::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) reply id=0x0c81, seq=1, hop limit=62
36	117.642247	2001:db8:c0de:11::a	2001:db8:c0de:13::a	ICMPv6	118	Echo (ping) request id=0x0c81, seq=2, hop limit=6
37	117.643815	2001:db8:c0de:13::a	2001:db8:c0de:11::a	ICMPv6	118	Echo (ping) reply id=0x0c81, seq=2, hop limit=62

Frame 33: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:04 (00:50:79:66:68:04), Dst: 0c:dc:1d:6f:00:02 (0c:dc:1d:6f:00:02)

Address Resolution Protocol (reply)

- Hardware type: Ethernet (1)
- Protocol type: IPv4 (0x0800)
- Hardware size: 6
- Protocol size: 4
- Opcode: reply (2)
- Sender MAC address: Private_66:68:04 (00:50:79:66:68:04)
- Sender IP address: 64.100.1.10
- Target MAC address: 0c:dc:1d:6f:00:02 (0c:dc:1d:6f:00:02)
- Target IP address: 64.100.1.1

Hex dump (0000-0030):

```

0000  0c dc 1d 6f 00 02 00 50 79 64 00 00 00 00 00 00
0010  08 00 06 04 00 02 00 50 79 64 00 00 00 00 00 00
0020  0c dc 1d 6f 00 02 40 64 01 00 00 00 00 00 00 00
0030  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
  
```

Рис. 8: Анализ трафика

Итоговая задача

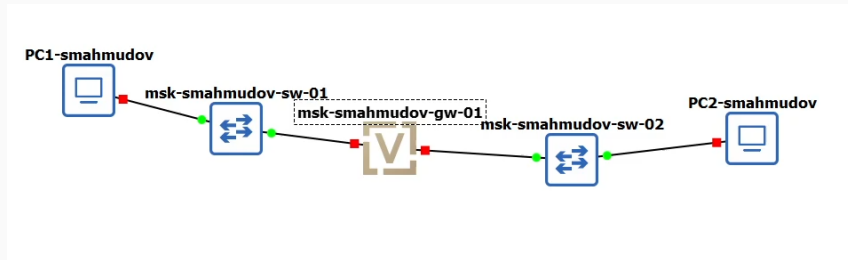


Рис. 9: Топология двух подсетей

```
VPCS> ping 10.10.1.20  
84 bytes from 10.10.1.20 icmp_seq=1 ttl=63 time=4.703 ms  
84 bytes from 10.10.1.20 icmp_seq=2 ttl=63 time=3.299 ms  
84 bytes from 10.10.1.20 icmp_seq=3 ttl=63 time=5.250 ms  
84 bytes from 10.10.1.20 icmp_seq=4 ttl=63 time=5.786 ms  
84 bytes from 10.10.1.20 icmp_seq=5 ttl=63 time=2.345 ms
```

Рис. 10: Проверка соединения

- Реализована сеть с двойным стеком IPv4/IPv6.
- Настроены маршрутизаторы **FRR** и **VyOS**.
- Проверена связность между всеми узлами.
- В **Wireshark** подтверждена корректная работа **ARP**, **ICMP** и **ICMPv6**.
- Полученные результаты демонстрируют правильную настройку адресации и взаимодействие устройств.